

## OPERATING SYSTEMS

### LAB SUBMISSION: 1 (BASIC LINUX COMMANDS)

This submission includes the application and results of basic Linux commands like: mkdir, rmdir, cd, ls command, grep command, pwd, cd, cp, mv, who, who am I, echo, chmod, exit and other frequently used commands.

```
student@student-VirtualBox:~$ mkdir tmp
student@student-VirtualBox:~$ mkdir memos letters e-mail
student@student-VirtualBox:~$ dir
24BAI1494  blueberrycopy  e-mail  fruits  Music  pt  test
a.out     Desktop        faculty Hello   new_pear Public tmp
apple     Documents      folder  letters orange snap  Videos
blueberry Downloads      folderr memos   Pictures Templates
student@student-VirtualBox:~$ rmdir letters
student@student-VirtualBox:~$ dir
24BAI1494  blueberrycopy  e-mail  fruits  new_pear Public tmp
a.out     Desktop        faculty Hello   orange snap  Videos
apple     Documents      folder  memos   Pictures Templates
blueberry Downloads      folderr Music  pt  test
student@student-VirtualBox:~$ cd 24BAI1494
student@student-VirtualBox:~/24BAI1494$ ls
Reneta
student@student-VirtualBox:~/24BAI1494$ cd
student@student-VirtualBox:~$ ls
24BAI1494  blueberrycopy  e-mail  fruits  new_pear Public tmp
a.out     Desktop        faculty Hello   orange snap  Videos
apple     Documents      folder  memos   Pictures Templates
blueberry Downloads      folderr Music  pt  test
student@student-VirtualBox:~$ grep c PatternList 23BRS1358
grep: PatternList: No such file or directory
grep: 23BRS1358: No such file or directory
student@student-VirtualBox:~$ grep c "hello" 23BRS1358
grep: hello: No such file or directory
grep: 23BRS1358: No such file or directory
student@student-VirtualBox:~$ grep c "is" faculty
grep: is: No such file or directory
grep: faculty: Is a directory
student@student-VirtualBox:~$ grep c "is" blueberrycopy
grep: is: No such file or directory
student@student-VirtualBox:~$ grep c "is" *
grep: is: No such file or directory
grep: 24BAI1378: Is a directory
grep: 24BAI1494: Is a directory
grep: a.out: binary file matches
grep: Desktop: Is a directory
grep: Documents: Is a directory
```

```
grep: fruits: Is a directory
grep: Hello: Is a directory
grep: memos: Is a directory
grep: Music: Is a directory
grep: new_pear: Is a directory
grep: Pictures: Is a directory
grep: pt: Is a directory
grep: Public: Is a directory
grep: snap: Is a directory
grep: Templates: Is a directory
grep: tmp: Is a directory
grep: Videos: Is a directory
student@student-VirtualBox:~$ pwd
/home/student
student@student-VirtualBox:~$ cat>copies.txt
cat>>copies.txt

hi,welcome to os lab
student@student-VirtualBox:~$ cp Videos fruits
cp: -r not specified; omitting directory 'Videos'
student@student-VirtualBox:~$ cp -r Videos fruits
student@student-VirtualBox:~$ mv faculty faculties
student@student-VirtualBox:~$ dir
24BAI1378  blueberry      Documents  folder  memos    Pictures  Templates
24BAI1494  blueberrycopy  Downloads  folderr Music    pt        test
a.out     copies.txt     e-mail    fruits  new_pear Public    tmp
apple     Desktop       faculties  Hello   orange   snap      Videos
student@student-VirtualBox:~$ who
student  seat0          2025-07-15 08:18 (login screen)
student  tty2           2025-07-15 08:18 (tty2)
student@student-VirtualBox:~$ who am i
student@student-VirtualBox:~$ dir
24BAI1378  blueberry      Documents  folder  memos    Pictures  Templates
24BAI1494  blueberrycopy  Downloads  folderr Music    pt        test
a.out     copies.txt     e-mail    fruits  new_pear Public    tmp
apple     Desktop       faculties  Hello   orange   snap      Videos
student@student-VirtualBox:~$ whoami
student
student@student-VirtualBox:~$ date
Tuesday 15 July 2025 09:13:32 AM IST
```

```
student@student-VirtualBox:~$ echo

student@student-VirtualBox:~$ head -n 2 apple
My name is Jiya
I am at os lab
student@student-VirtualBox:~$ tail -n 1 apple
I am studying basic commands
student@student-VirtualBox:~$ find. -iname "apple"
Command 'find.' not found, did you mean:
  command 'findv' from deb polylib-utils (5.22.5-4+dfsg)
  command 'find' from deb findutils (4.9.0-5)
  command 'findg' from deb ncl-ncarg (6.6.2.dfsg.1-3)
Try: sudo apt install <deb name>
student@student-VirtualBox:~$ find -iname "apple"
./apple
student@student-VirtualBox:~$ chmod a+r apple
student@student-VirtualBox:~$ man
What manual page do you want?
For example, try 'man man'.
student@student-VirtualBox:~$ man man
man: can't resolve man7/groff_man.7
student@student-VirtualBox:~$ history
  1  sudo apt update && sudo apt upgrade -y
  2  sudo apt install ruby \full
  3  sudo apt install ruby
  4  cat >hello.rb
  5  cat> hello.rb
  6  ruby.rb
  7  ruby hello.rb
  8  ruby -v
  9  lsb_release -a
 10  clear
 11  exit
 12  ruby -v
 13  ./ autorun.sh
 14  ./autorun.sh
 15  asdf install ruby 3.3.3
 16  sudo apt install asdf
 17  sudo apt install rails
 18  ruby -v
```

```
19 clear
20 ruby hello.rb
21 sudo apt update
22 sudo apt install build-essential
23 gcc --version
24 vi first.c
25 gcc first.c
26 ./a.out
27 sudo apt install net-tool
28 ifconfig
29 sudo apt install net-tools
30 ifconfig
31 cd ~
32 touch mid.c
33 nano mid.c
34 gcc mid.c -o mid
35 nano mid.c
36 gcc reads.c
37 touch reads.c
38 gcc read.c
39 ./a.out
40 gcc read.c
41 ./a.out
42 cat /etc/passwd \ more
43 cat /etc/passwd
44 ls
45 gcc -version
46 gcc --version
47 mpi
48 mpic
49 mpic --version
50 mpi --version
51 mpicc
52 sudo apt install lam4-dev
53 sudo apt install libhdf5-mpich-dev
54 sudo apt install libhdf5-openmpi-dev
55 sudo apt install mpich
56 sudo apt install openmpi-bin~
57 sudo apt install openmpi-bin
58 mpich
```

```
252 cp videos fruits
253 cp -r Videos fruits
254 mv faculty faculties
255 dir
256 who
257 who am i
258 dir
259 whoami
260 date
261 echo
262 head -n 2 apple
263 tail -n 1 apple
264 find. -iname "apple"
265 find -iname "apple"
266 chmod a+r apple
267 man
268 man man
269 history
student@student-VirtualBox:~$ more FILE
more: cannot open FILE: No such file or directory
student@student-VirtualBox:~$ more apple
My name is Jiya
I am at os lab
I am studying basic commands
student@student-VirtualBox:~$
```